SecAI+ Lab 1

How AI models learn (and fail)

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Part A: Results

Spiral dataset; X1 and X2; Tanh; learning rate 0.03; batch size 10; no regularization. Seeds: 0.25258 (default and larger); 0.38080 (noisy).

Run	Layers	Noise	Train %	Epochs	Train loss	Test loss
Default	4, 2	0	50%	325	0.433	0.486
Larger	8, 8, 8	0	50%	1080	0.005	0.016
Noisy	8, 8, 8	50	10%	780	0.303	0.611

What happened in step A4?

With noise at 50 and training data at 10%, training loss was 0.303 and test loss was 0.611. The model fit the training examples better than the unseen test examples. This gap is evidence of overfitting: the model learned details that did not apply well to new data.

Why does data poisoning concern a defender?

An attacker can add false examples or incorrect labels to the training data. The model can learn these errors and miss attacks or flag safe activity as a threat. This attack is called data poisoning.

Part B: Tokens

OpenAI Tokenizer | Selected model group: GPT-5.x & O1/3. Inputs have no leading spaces or final newline.

Type	Exact input	Tokens	Chars
English	The model learns from training data.	7	36
Python	print("Hello, world!")	6	22
Example string	Gr@mbl1ng!2026	8	14

Rare strings split into more tokens because the tokenizer represents unfamiliar character sequences with smaller pieces from its vocabulary.

Sources: playground.tensorflow.org; platform.openai.com/tokenizer.

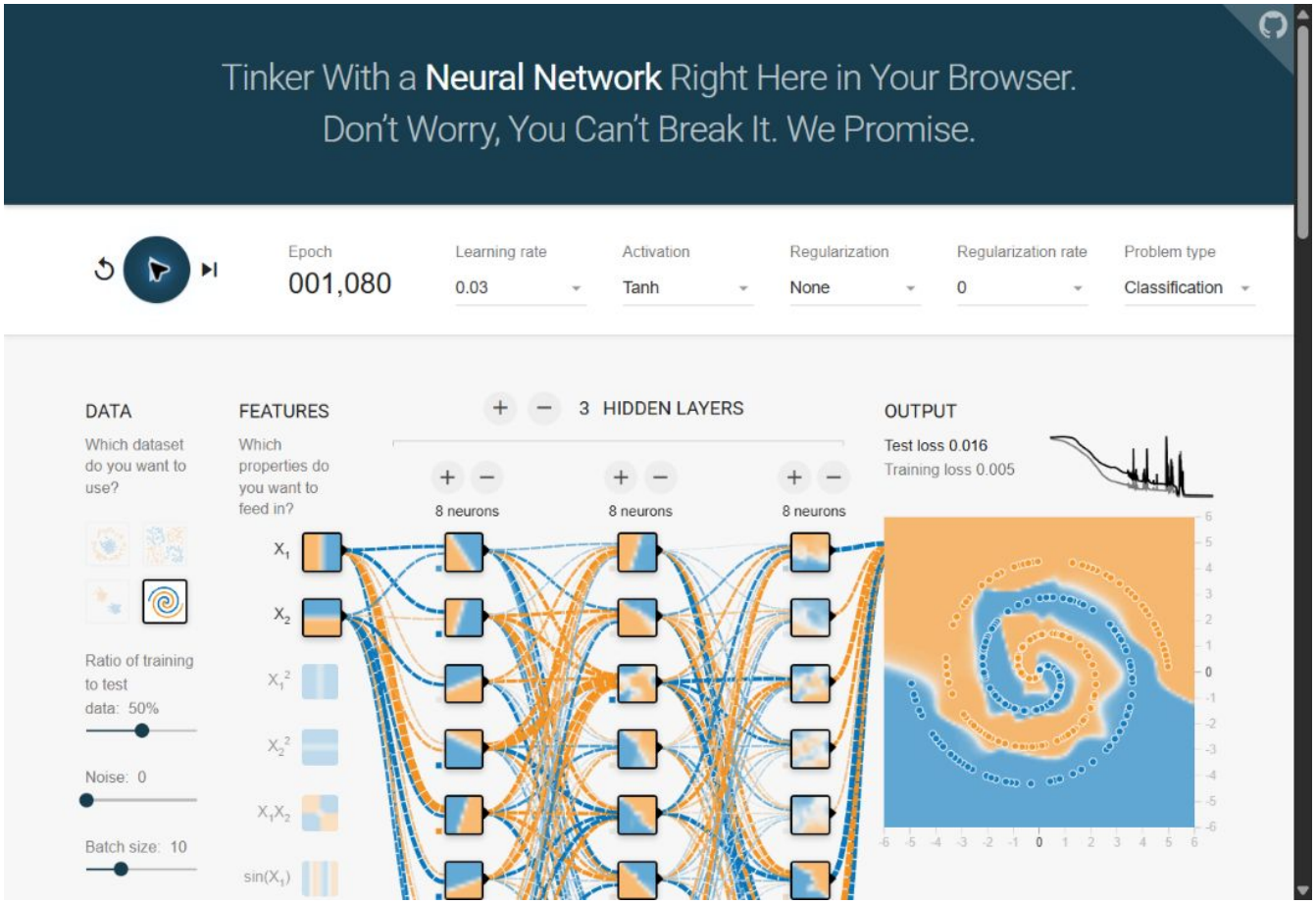
Default network

325 epochs. Hidden layers: 4, 2. Training loss: 0.433. Test loss: 0.486. The network did not separate the spiral well.



Larger network

1080 epochs. Hidden layers: 8, 8, 8. Training loss: 0.005. Test loss: 0.016. The output follows the spiral well. This run also had more training time.



Noise and limited training data

780 epochs. Hidden layers: 8, 8, 8. Noise: 50. Training data: 10%. Training loss: 0.303. Test loss: 0.611.

